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India's Shipbuilding Ambitions Can Set Sail With Korea

India's shipbuilding ambitions can set sail with Korea

In April this year, South Korean President Lee Jae Myung met Prime Minister Narendra Modi in India, his first visit to India and the first by a Korean leader in eight years. The meeting was remarkable in many ways, as it revived high-level political interaction at the leaders' level, resulting in the expansion of partnership in strategic sectors. However, the visit was special for another reason: it paved the way for strong collaboration between India and South Korea in the shipbuilding industry, a strategic sector that India is seeking to revive.

Seoul plugs into India's shipbuilding drive
Mr. Lee's visit has given positive momentum to the slowly progressing India-South Korea shipbuilding partnership, as seen in the slew of memoranda of understanding (MoU) and agreements signed during his trip. While the big three of the South Korean shipbuilding industry – Samsung Heavy Industries (SHI), HD Korea Shipbuilding & Offshore Engineering, and Hanwha Ocean – had already announced their investment plans, partnerships, or interests in India, Mr. Lee's support reiterates the South Korean government's strategic commitment to shipbuilding collaboration with India. Last year, a subsidiary of Hyundai signed an MoU with Cochin Shipyard Limited and has since announced plans to invest \$4 billion in Thoothukudi, Tamil Nadu, to construct a green shipyard. Similarly, SHI has signed a partnership with Swan Defence and Heavy Industries to build ships in India. These developments showcase India's attractiveness as a shipbuilding destination for global giants.

South Korea is also interested in developing a complementary supply chain that includes ancillary industries. For instance, the Korea Marine Equipment Association (KOMEA), which comprises 304 enterprises across ship design, shipbuilding, marine equipment, and ship repair, has opened an office in Mumbai. This is expected to pave the way for the development of a robust shipbuilding ecosystem in India. Such steps will help foster a shipbuilding cluster encompassing ancillary industries and other critical facilities



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South Korean partnerships could help accelerate India's ambitions to become a shipbuilding powerhouse

required for scaling up the sector. Further, these partnerships are expected to provide the Indian shipbuilding industry with much-needed support, including design and engineering expertise as well as production know-how. This will help India develop human capital and adopt technology at scale, making its shipyards more competitive internationally. In addition, the various MoUs signed between India and South Korea – at both the government-to-government and business-to-business levels – on workforce development, maritime education, research, and innovation will further strengthen the shipbuilding ecosystem. Together, these efforts could help establish a holistic, cluster-led development model inspired by the success of the city of Ulsan in South Korea.

What has been accomplished in such a short time is commendable, but the task is far from complete. To emerge as a leading shipbuilding nation, India will have to pursue multiple objectives simultaneously. It must support the industry through proactive policy and fiscal measures while also responding effectively to external crises that could disrupt supply chains and affect demand. The sector will require sustained hand-holding until it becomes self-sufficient and capable of competing globally, particularly with established giants such as China.

Fill the gaps

There must be a focus on human capital development, policy and fiscal support, and bringing in ancillary industries. India's Maritime Vision 2030 and Maritime Amrit Kaal Vision 2047 clearly state the objective of being among the top 10 shipbuilding nations by 2030 and in the top five by 2047.

Complementing initiatives such as the Maritime Development Fund, Shipbuilding Development Scheme, and Shipbuilding Financial Assistance Policy make it clearer that India is serious about attracting foreign investment in the shipping and shipbuilding sector.

However, policy and operational gaps persist.

To rectify these gaps, India will have to focus on implementing a series of reforms related to regulatory consistency and legal predictability and also providing access to low-cost and long-term capital. Steps such as the creation of the Sagarmala Finance Corporation Limited (SFCL), India's first non-banking financial company for the maritime sector, are a positive and welcome development.

However, the greater challenge will be establishing a comprehensive industrial ecosystem for shipbuilding. To achieve this, India will need to move quickly on workforce development, supplier localisation, and the creation of dedicated maritime institutions. In addition, Indian academia and research institutions will have to play a larger role in this developmental partnership to support the country's shipbuilding ambitions.

India must continue to focus on three key priorities: providing sustained policy and fiscal support, developing the capacity to absorb transferred technologies, and formulating a sectoral strategy with clearly defined goals and targets. The in-principle approval of the greenfield project in Tamil Nadu is a welcome development, signalling that approval and implementation bottlenecks need not be insurmountable. To capitalise on this opportunity, however, State governments, alongside the central government, must ensure timely follow-through at every stage, facilitate the entry of foreign investors, and provide continuous support throughout the investment process.

A proven pathway

While India's shipbuilding ambitions are ambitious, they are not impractical. South Korea's shipbuilding journey, from a minor player to a global leader in just 15 years, beginning in the 1970s, demonstrates what is possible. India can replicate that success by focusing on three priorities: sustained policy and financial support, competitive shipbuilding and industrial capacity, and a skilled workforce.

India's Shipbuilding Ambitions Can Set Sail With Korea

● The Trigger

- In April 2026, South Korean President Lee Jae Myung met PM Modi in India, his first visit to India, and the first by a Korean leader in 8 years

● India's Targets & Existing Schemes

- Maritime Vision 2030 and Maritime Amrit Kaal Vision 2047 set goals of being among the top 10 shipbuilding nations by 2030 and in the top 5 by 2047
- Existing initiatives: Maritime Development Fund, Shipbuilding Development Scheme, Shipbuilding Financial Assistance Policy, and the Sagarmala Finance Corporation Limited (SFCL) : India's first NBFC dedicated to the maritime sector

● Gaps and the Way Forward

- Persistent gaps: Regulatory consistency, Legal predictability, and access to low-cost, long-term capital
- Need to focus on workforce development, supplier localisation, and dedicated maritime institutions; academia/research institutions must play a larger role
- South Korea's own journey : from a minor player to a global shipbuilding leader in just 15 years, beginning in the 1970s shows India's ambitions, while challenging, are not impractical

India's Shipbuilding Ambitions Can Set Sail With Korea

Global Standing: India currently holds less than 1% of the global shipbuilding market share, significantly trailing leaders like China (47%), South Korea (25%), and Japan (18%).

Challenges

High Capital Costs: Indian shipyards face borrowing costs of 9–11%, compared to 4–6% in China and South Korea, leading to a 20% cost disadvantage.

Fragmented Ecosystem: Lack of integrated clusters means shipyards are far from component suppliers, increasing logistics costs and timelines.

Import Dependency: India still imports 60–70% of critical marine components, including specialized steel, propulsion engines, and navigation electronics.

Market Preference: Indian ship-owners often prefer buying second-hand foreign vessels due to lower costs and immediate availability.

Underdeveloped MRO (Repair): Most Indian-flagged vessels prefer undergoing maintenance in Singapore, Dubai, or Colombo due to superior infrastructure and quicker turnaround.

India's Shipbuilding Ambitions Can Set Sail With Korea

Major Ports are owned and managed by the Central Government while Minor or other Non-Major Ports are owned and managed by the State Governments.

Major Ports are under the Union List of the Constitution of India while the Non-Major Ports are under the Concurrent List.

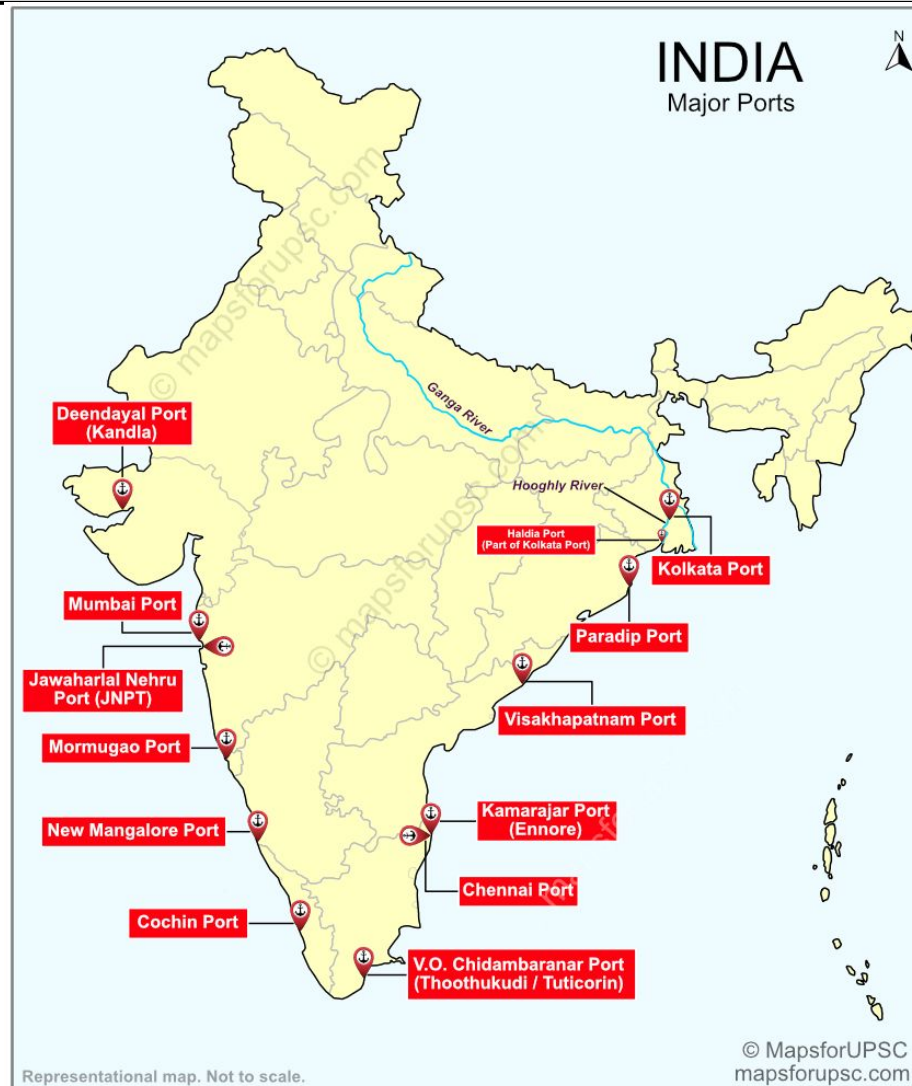
There are 12 Major Ports in India out of which 6 are located on the East Coast and 6 on the West Coast.

Galathea Bay Port (proposed 13th Major Port of India):

The Galathea Bay Port (also known as the International Container Transshipment Port (ICTP)) is a proposed container transshipment port at Galathea Bay, Great Nicobar Island in the union territory of Andaman and Nicobar Islands

In September 2024, it was officially notified as the 13th major port of India.

India's Shipbuilding Ambitions Can Set Sail With Korea



India's Shipbuilding Ambitions Can Set Sail With Korea

1. Deendayal Port (Kandla Port)

Located in Kutch District, the Deendayal (Kandla) Port is the largest port in India by volume of cargo handled.

2. Mumbai Port : **It is a natural deep-water port.** It is the **largest port in India in terms of size.** It is also known as **Nhava Sheva.** It is the **largest container port (among major ports) in India, handling around 50% of the total containers handled by all Major Ports in India.** The **Mundra Port is also India's first private port** and largest commercial port, located on the northern

3. JNPT Port : It is also known as Nhava Sheva. It is the largest container port (among major ports) in India, handling around 50% of the total containers handled by all Major Ports in India.

4. Mormugao Port

Located in the state of Goa, the Mormugao Port has a naturally protected open-type harbour. It lies on the mouth of the river Zuari. It is the premier iron-ore exporting port.

5. News Mangalore Project

6. Cochin Port : It lies on two islands in the Vembanad Lake : Willingdon Island and Vallarpadam, towards the Fort Kochi river mouth opening onto the Laccadive Sea.

7. V. O. Chidambaranar Port (Tuticorin Port) : It is an artificial, all-weather port located in Gulf of Mannar, Tamil Nadu.

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8. Chennai Port

It is the largest and oldest artificial harbour in India.

9. Kamarajar Port (Ennore Port)

Located on the Coromandel Coast just about 20 km north of the Chennai Port.

It is the first port in India which is a public company.

10. Visakhapatnam Port

It is the deepest natural harbour of India.

11. Paradip Port

It is situated on the confluence of the Mahanadi River and the Bay of Bengal in Odisha.

It is a natural, deep-water port.

It is the key port for the export of iron ore to Japan.

12. The Syama Prasad Mookerjee Port (Kolkata Port)

Established in 1870, It is the oldest operating port in India.

It is the only riverine Major Port in India.

India's Shipbuilding Ambitions Can Set Sail With Korea

Consider the following pairs :

	Port	Well known as
(1)	Kamarajar Port	First major port in India registered as a company
(2)	Mundra Port	Largest privately owned port in India
(3)	Visakhapatnam Port	Largest container port in India

How many of the above pairs are correctly matched?

This question was previously asked in

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1. Only one pair
2. Only two pairs
3. All three pairs
4. None of the pairs

Revisiting India's Ultrasound Laws

Revisiting India's ultrasound laws

Technological shifts and rising need for community-based cancer screening suggest that regulatory frameworks should adapt to distinguish between different use cases of ultrasound. An amendment to the PCPNDT Act could legalise community-based ultrasound using a high-frequency linear probe, as it will not impact sex determination

FULL CONTEXT

Parth Sharma
Senthil Kumar A.R.

Mrs. Janki (name changed), a 45-year-old woman, came to a health camp organised in a rural Assam with a complaint of a painless breast lump. She had noticed the lump three months earlier and reported that it had gradually increased in size. On examination, the doctor suspected breast cancer and advised her to visit a cancer hospital located two hours away for further evaluation. "But I have no pain. The hospital is so far away. There is nobody to come with me to the hospital," Mrs. Janki said. Despite repeated counselling, she refused to seek further care.

Three months later, Janki finally presented to the hospital. By then, she had developed a large, bleeding mass in her breast. She was diagnosed with advanced stage breast cancer and died six months later.

Technological advances, the portable ultrasound machines, have made it possible to bring cancer diagnostics, such as ultrasound imaging and ultrasound-guided biopsies, closer to people's homes. This machine could, perhaps, have given Mrs. Janki a diagnosis at her doorstep. However, India's Pre-Conception and Prenatal Diagnostic Techniques (PCPNDT) Act treats the movement of an ultrasound machine outside a registered facility as a serious offence, with penalties that can include a minimum of three months non-bailable imprisonment.

Provisions of the PCPNDT Act

The PCPNDT Act was introduced in 1994 in response to a sharp decline in the child sex ratio in India, driven by a strong societal preference for male children and the growing misuse of technologies such as ultrasonography for prenatal sex determination, followed by selective abortion of female foetuses. Abortion of female foetuses became more evident from the 1980s, with the increasing availability of imaging technologies, raising serious ethical, demographic, and public health concerns. The legislation, therefore, aimed not only to prevent the misuse of medical technology but also to address the deeper issue of gender discrimination.

The law mandates registration of all genetic clinics, ultrasound centres, and laboratories and strictly prohibits the communication or disclosure of the sex of the foetus. It prescribes detailed record-keeping, monitoring mechanisms, and penalties to ensure compliance.

Under this Act, the purchase of an ultrasound machine is not a routine commercial transaction but a tightly-regulated process. A clinic or hospital must first be registered with the district government as a genetic clinic or imaging centre before acquiring a machine. Purchasing one without prior registration is illegal. Manufacturers or dealers are required to verify the buyer's credentials and obtain a written undertaking that the machine will not be used for sex determination. The sale must be documented with invoice details that match the registered centre, and the transaction is reported to the authorities for monitoring.

Once installed, the machine must remain at the approved location. Its details must be recorded in official registers, and strict documentation,



Expanding access: Portable, handheld ultrasound devices can make it technically feasible to bring diagnostic services closer to patients' homes. GETTY IMAGES

including patient records for every scan, must be maintained.

Impact of PCPNDT Act

Following the introduction of the PCPNDT Act, the sex ratio at birth in India has shown a gradual improvement at the national level. While this improvement cannot be conclusively attributed to the introduction of the Act, several unintended consequences can be directly attributed to it.

Evidence suggests that families whose first child is a girl tend to have more children in an effort to have a son, compared to families with a firstborn boy. Following restrictions on prenatal sex selection, families experienced a 25% higher child mortality rate among firstborn girls relative to firstborn boys. This could be explained by the decline in parental investment in healthy, particularly among girls. Fertility also increased in these families, indicating a shift toward having more children when sex selection was not an option. Having a larger number of children also led to dilution of resources, likely reducing overall investment per child. Similar patterns were observed in education as gender disparities widened. These effects were more pronounced among poorer, rural households that were more affected by the restrictions, as they could not afford legal services to abort the child.

Recent reports indicate that sex-selective practices continue despite decades of legal prohibition. In October 2025, authorities uncovered an illegal racket in Karnataka where organised networks were carrying out prenatal sex determination and facilitating abortions, often targeting vulnerable women from rural areas. Similar crackdowns across different States show that such activities are frequently conducted outside formal health systems, using portable ultrasound devices, informal providers, and covert arrangements to evade regulation.

The issue is not confined to India. Reports from the United Kingdom indicate that sex preference may persist among some diaspora communities as well. Recent research has also demonstrated that families of Indian origin may be engaging in sex-selective practices even in

settings with stricter oversight, highlighting the deep-rooted nature of gender bias that transcends national boundaries.

Taken together, these patterns underscore a critical limitation of legislative approaches. While laws such as the PCPNDT Act establish an essential regulatory framework, they are insufficient on their own to drive social change.

The case for reform

Available evidence suggests that the PCPNDT Act has not fully achieved its intended objectives and may, in some contexts, have had unintended adverse effects. Additionally, fear of legal repercussions and stringent regulatory requirements has had a chilling effect on service provision.

Moreover, the PCPNDT Act has not kept pace with advances in ultrasound technology. Portable, handheld ultrasound devices, often connected to smartphones or tablets, make it technically feasible to bring diagnostic services closer to patients' homes, which is particularly relevant for early cancer detection in underserved areas. However, the use of such devices at the community level is currently illegal in India.

Modern high-frequency probes used for applications such as cancer detection and the assessment of other superficial conditions cannot be used for foetal sex determination, yet remain subject to the same regulatory restrictions, limiting access to essential diagnostic care.

Recent developments in artificial intelligence (AI) further strengthen the case for a more nuanced regulatory approach. AI-enabled ultrasound systems can assist with image acquisition and interpretation, and in some configurations generate automated reports based on pattern recognition without requiring full image storage or display. Such technologies could enable safe, purpose-specific use of ultrasound for diagnostic applications while substantially reducing the risk of misuse for foetal sex determination.

Recent research has also demonstrated the potential of AI in improving access to reliable ultrasonography. In a pilot study,

portable ultrasound scans performed by individuals with minimal training were combined with AI to identify suspicious breast lesions with high accuracy, correctly flagging all confirmed cancer cases. This suggests that AI-assisted ultrasound could enable frontline health workers to assess patients with breast lumps, refer those with suspicious findings for further evaluation, and reassure those with benign conditions.

For countries such as India, where access to specialist radiologists and diagnostic imaging remains uneven, especially in rural areas where nearly 70% of the population resides, such innovations could substantially improve the timely assessment and treatment of symptomatic patients. Earlier diagnosis and referral have the potential to reduce breast cancer mortality while expanding access to care in rural and underserved areas. This contrasts with many Western countries, where breast cancer control strategies rely more heavily on mammographic screening programmes that detect tumours before symptoms appear but require considerably greater resources and infrastructure.

Serving a greater need

These technological shifts and the rising need for community-based cancer screening in view of the high cancer burden in India suggest that regulatory frameworks need to adapt to distinguish between different use cases of ultrasound. An amendment to the Act could make community-based ultrasound using a high-frequency linear probe legal, as it will not impact sex determination. Additionally, the Act should incorporate provisions addressing emerging technologies, including AI-enabled and safeguarded USG imaging systems designed to prevent the determination or disclosure of foetal sex, irrespective of intent.

Dr Parth Sharma is a community physician working in the department of preventive oncology and public health at Asha Cancer Hospital and Research Centre. Dr Senthil Kumar A.R. is a consultant radiologist and former director of the department of radiology, Aberdeen Royal Infirmary, UK. Views are personal.

THE GIST

▼ The PCPNDT Act was introduced in 1994 to curb prenatal sex determination and the selective abortion of female foetuses in response to a declining child sex ratio.

▼ The law has not kept pace with advances in ultrasound technology, including portable handheld devices that can bring diagnostic services closer to patients' homes.

▼ For countries such as India, where access to specialist radiologists and diagnostic imaging remains uneven, such innovation could aid early cancer detection, particularly in underserved areas.

Revisiting India's Ultrasound Laws

- **The PCPNDT Act**

- The **Pre-Conception and Pre-Natal Diagnostic Techniques (PCPNDT) Act, 1994** was introduced in response to a sharp decline in India's child sex ratio, driven by **misuse of ultrasonography for prenatal sex determination and selective abortion of female fetuses**
- **Mandates registration of all genetic clinics, ultrasound centres and laboratories; strictly prohibits communication/disclosure of foetal sex; prescribes detailed record-keeping and monitoring**
- **Treats movement of an ultrasound machine outside a registered facility as a serious offence, with penalties including a minimum of 3 months' non-bailable imprisonment**

Revisiting India's Ultrasound Laws

● The Case for Reform

- The law hasn't kept pace with advances in ultrasound technology **portable, handheld devices could bring diagnostic services** like cancer screening closer to patients' homes, especially in underserved rural areas (where nearly 70% of India's population lives)
- AI-enabled ultrasound systems can assist image acquisition/interpretation and generate automated reports via pattern recognition, without needing full image storage potentially enabling safe, purpose-specific diagnostic use

● The Proposed Amendment

- The Act should also incorporate provisions for emerging technologies, including AI-enabled and safeguarded ultrasound imaging systems designed to prevent determination/disclosure of foetal sex, irrespective of intent

A Nalanda Model for the 21st Century



SYED ATA
HASNAIN

It represented pluralism and openness to ideas — qualities increasingly important in a world seeking to balance technological advancement with ethical reflection

A Nalanda model for the 21st century

THE GREAT contests of the 21st century are unlikely to be decided solely on battlefields, in factories or through financial markets. Nations today are competing for intellectual advantage — the ability to generate ideas, attract talent, shape narratives and influence the ethical direction of technology and governance. Knowledge is becoming national power.

History shows that every era has had a defining source of strategic advantage. Territorial conquest shaped the age of empire. The Industrial Revolution transformed Britain, Germany and later the US. The Information Revolution rewarded societies that mastered computing, telecommunications and digital systems. The world is now entering the age of cognitive power.

This extends far beyond AI. Cognitive power includes scientific research, climate knowledge, disaster management, biotechnology, cyber capability, strategic communication, conflict studies, diplomacy and the ability to shape international norms. The present era is distinguished by the unprecedented strategic value attached to knowledge, research and intellectual capital.

The US understood this transformation earlier than most. Its advantage lay not merely in great universities but in linking academia, industry, government funding, military research and immigration into a single innovation ecosystem. Institutions like MIT, Stanford and Harvard became strategic assets connected directly to aerospace, semiconductors, biotechnology and digital innovation.

Equally important was academic freedom. Universities retained sufficient autonomy to encourage experimentation and dissent, allowing talent from around the world to become strategic capital. Innovation rarely flourishes in climates of intimidation. This lesson has universal relevance.

India adapted impressively during the Information Revolution. Yet its role remained concentrated more in software services than in foundational research. The next leap, therefore, cannot merely be technological. India must move from being a digital economy to becoming a knowledge civilisation.

Nations that invest solely in coding and

automation without nurturing ethics, humanities, law and strategic thought may achieve efficiency without intellectual depth. History reminds us that technological breakthroughs acquire real significance only when accompanied by the ideas and doctrines that give them purpose.

Knowledge power is increasingly central to national resilience and security. Nations that anticipate crises, shape narratives and build resilience will enjoy geopolitical advantage. India's CDRI is a welcome step. Future conflicts will require strategic thinkers and diplomats as much as technologists.

Encouragingly, India is beginning to recognise the relationship between knowledge ecosystems and long-term strategic development. Economic modernisation can no longer rely only on infrastructure and industrial expansion. Future competitiveness will depend equally upon research, innovation and higher education.

In this context, Nalanda in Bihar acquires renewed relevance. More than a millennium ago, it functioned as a transnational intellectual network attracting scholars from across Asia. It represented pluralism, interdisciplinary learning and openness to ideas — qualities increasingly important in a world seeking to balance technological advancement with ethical reflection. Properly nurtured, modern Nalanda could emerge as a centre for Asian cooperation in culture, sustainability, strategic studies and global governance.

The strategic reality of the 21st century is becoming increasingly clear. Military strength will continue to matter, but military capability itself will increasingly depend upon research ecosystems, innovation cultures and intellectual capital.

Nations unable to generate original knowledge will become dependent on others not only for technology, but also for strategic narratives and policy frameworks. The emerging age of cognitive power offers India an opportunity to become once again a civilisation that produces ideas, institutions and knowledge systems sought by the world.

The writer is Governor, Bihar and former commander of the Indian Army's Srinagar-based Chinara Corps

A Nalanda Model for the 21st Century

● The Core Argument

- 21st-century power contests will be decided less by battlefields, factories or financial markets, and more by intellectual advantage the ability to generate ideas, attract talent, shape narratives and influence the ethical direction of technology and governance
- Knowledge is becoming national power; **the world is entering an “age of cognitive power”**

● Historical Pattern of Strategic Advantage

- **Each era has had its defining source of advantage**: territorial conquest (age of empire) → Industrial Revolution (Britain, Germany, later the U.S.) → Information Revolution (computing, telecommunications, digital systems)
- **Cognitive power extends beyond AI**: it includes scientific research, climate knowledge, disaster management, biotechnology, cyber capability, strategic communication, conflict studies, diplomacy, and the ability to shape international norms

A Nalanda Model for the 21st Century

The U.S. Model

- The U.S.'s advantage lay not merely in great universities, but in linking academia, industry, government funding, military research and immigration into a single innovation ecosystem (MIT, Stanford, Harvard as strategic assets connected to aerospace, semiconductors, biotech, digital innovation)
- Academic freedom was equally important universities retained autonomy to encourage experimentation and dissent, since innovation rarely flourishes under intimidation

● India's Position

- India adapted well during the Information Revolution, but its role stayed concentrated in software services rather than foundational research
- Nations that invest only in coding and automation without nurturing ethics, humanities, law and strategic thought may achieve efficiency without intellectual depth

● Reviving Nalanda

- Nalanda in Bihar, over a millennium ago, functioned as a **transnational intellectual network attracting scholars from across Asia representing pluralism, interdisciplinary learning and openness to ideas**
- Properly nurtured, a modern Nalanda model could position India as a centre for Asian cooperation in culture, sustainability, strategic studies and global governance
- Nations unable to generate original knowledge will become dependent on others not just for technology, but for strategic narratives and policy frameworks

Keeping Humanity at the Centre of the AI Revolution

Keeping humanity at the centre of the AI revolution

At the peak of human history, the Artificial Intelligence (AI) revolution stands out as a transformative moment, harnessing the infinite potential of science and innovation in the service of humankind. The endless possibilities of AI in every sphere of human activity and its empowering potential to deal with the unprecedented challenges of our times, is a tribute to mankind's collective ingenuity that validates the boast of accomplishments of which 'even the gods might be envious'. The automating of tedious tasks involving repetitive work and freeing time for leisure, expanding access to essential services, breakthroughs in medicine, ensuring longevity and better health care including cancer screening and prediction of terminal illnesses, robotic nursing of the sick, a more effective targeting of economic aid to the marginalised, enhanced accessibility of education and knowledge to all and its vast contribution to environmental sustainability in several ways including disaster management and weather forecasting are some of the crowning contributions of AI toward a more inclusive development agenda.

Technology and human values

Even so, questions about the future of human society and the 'destiny of intelligence' are most vigorously debated by philosophers, scientists, statesmen and the technology czars, some of whom have pleaded for a pause in the further development and deployment of AI that could hijack our idea of humanity. The claims of the AI protagonists about the 'amazing abundance' of goods and services and the caution against slowing down of scientific progress through regulation must contend with the compelling questions about the necessity of ethical guardrails warranted by humanitarian considerations and emotions embedded in the deepest recesses of our being through a cultural evolution spanning the millennia. The question of who we are, and whether we are ready for a new narrative of humanity in which functional efficiency and promised material abundance prevail over the yearnings of the human soul and the dignity of emotion, is an unavoidable and larger inquiry that has resisted the seduction of technological wonders. This is particularly important given that AI is able to replicate, and in some cases even outperform, cognitive skills, including the ability to understand human emotions and intuition.

Are we ready for an endless technological disruption and a 'global epidemic of stress' caused by a prolonged volatility in the job market



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and a predicted 'useless' class of millions caused by an unmanageable stress in coping with the effects of technological disruption? Whether we have new social and economic models that protect individual self-worth and ensure for all a life of belonging and emotional well-being are larger philosophical questions arising from a humanist perspective founded on the sacrosanctity of the values we carry as a badge of humanity.

Clearly, the fact that passionate propounders of AI have cautioned against 'summoning the devil without a kill switch', must awaken us to the urgency of serious reflection on how to preserve 'the hidden realm of the mind from which emotions emerge, from which inspiration flows, from which our desires pulse – the subjective part of the human spirit that makes each of us ineluctably who we are'.

Data privacy vulnerabilities, proliferation of misinformation, electoral manipulations, the possibility of super intelligent weapons systems going rogue, AI-enabled phishing campaigns, and surveillance and censorship are some of the ominous portents that can cause social upheavals without effective global regulation.

Preserving the digital sovereignty of nation-states remains a challenge, given that control over data is intrinsically linked to national security and the strategic autonomy of nations. The establishment of a global regulatory regime that respects national sovereignty and ensures effective enforcement can no longer be postponed.

A moral compass

The world of AI that could rewrite the code of humanity, therefore, needs a moral vision that can harmonise technological advancement with the preconditions of a virtuous and happy society. Terry Eagleton's caution in relation to a sense of seductive self-assurance is eloquent and I quote: 'An inflated self-belief can earn its calamitous comeuppance, which caused the ancient Greeks to shudder and look fearfully to the skies'.

Spanish philosopher José Ortega y Gasset reminded us that 'we live at a time when man believes himself fabulously capable of creation, but he does not know what to create. Lord of all things, he is not Lord of himself. He feels lost amid his own abundance. With more means at its disposal, more knowledge, more technique than ever, it turns out that the world today goes the same way as the worst of worlds that have been; it simply drifts'.

The encyclical letter of His Holiness, Pope Leo

XIV on 'Safeguarding the Human Person in the Time of Artificial Intelligence', clinches the moral debate on the humanist dilemmas in relation to AI. Recognising that technology is not antagonistic to humanity, the Pope has stressed the duty to remain 'profoundly human' in an era of AI when 'human dignity is threatened by new forms of dehumanization'.

Emphasising the need to establish standards of ethical discernment based on the dignity of the individual, His Holiness has warned against the 'illusion' of 'self assertion' and cautioned against progress that exacerbates inequalities and is incapable of healing peoples' wounds. He has called for a rejection of the 'idolatry of profit that sacrifices the weak, a uniformity that neutralizes differences and the pretense that a single language – even a digital one – can translate everything, including the mystery of the person into data and performance'. These assertions by the Pontiff are moral injunctions in a world that needed a reminder of the 'splendour and grandeur of humanity' beyond its creations and of the limitations of AI in terms of its 'affective, relational or spiritual capabilities'.

The moral code propounded by the Pope, a product of an encounter between lived experiences, spiritual consciousness and historical memory, asserts that humanity flourishes in the 'fragility and finitude' of the human person and in its limitations.

Ethical AI governance

Thus viewed in the framework of core human values anchored in the dignity of man, global leaders are expected to opt for a 'humanist centric' approach in the deployment and regulation of AI for the larger good of humanity with the individual at the centre of their decisions. Prime Minister Narendra Modi has stressed at both the VivaTech 2026 conference in Paris (June 2026) and the India-AI Impact Summit 2026 in New Delhi (February 2026) that such an approach would require a robust and enforceable regulatory framework, rather than voluntary and non-binding commitments, to govern the use of AI. Such a framework could democratise access to frontier AI and help build a shared, trustworthy AI ecosystem in an age that 'levels everything and reveres nothing'. How nations deal with this epochal challenge will define the quality of political leadership and our commitment to inclusive democracy anchored in equality and human dignity as the ultimate civilisational aspiration.

The views expressed are personal

AI's promise must be guided by ethics, dignity and accountability

Keeping Humanity at the Centre of the AI Revolution

● AI's Transformative Promise

○ AI is automating tedious/repetitive tasks, expanding access to essential services, enabling breakthroughs in medicine (cancer screening, terminal illness prediction), robotic nursing care, more effective targeting of economic aid to the marginalised, and contributing to environmental sustainability (disaster management, weather forecasting)

● The Underlying Tension

○ Concerns include a predicted “useless” class of millions displaced by automation, data privacy vulnerabilities, misinformation, electoral manipulation, the risk of “superintelligent” weapons systems going rogue, AI-enabled phishing, and surveillance/censorship

○ Preserving the digital sovereignty of nation-states is a related challenge, since control over data is intrinsically linked to national security and strategic autonomy

● Towards Ethical AI Governance

○ Global leaders are urged to adopt a “humanist-centric” approach to AI deployment and regulation, keeping the individual at the centre of decisions

Netra, India's Airborne Surveillance Platform, Gets Final Operational Nod

Netra, India's airborne surveillance platform, gets final operational nod

Hemanth C.S.
BENGALURU

The indigenous Netra Airborne Early Warning and Control (AEW&C) system, which played a pivotal role as a force multiplier for the Indian Air Force in the 2019 Balakot strikes and Operation Sindoor last year, was accorded final operational clearance (FOC) on Thursday.

Developed indigenously by the Bengaluru-based Centre for Airborne Systems (CABS), the Netra AEW&C is integrated on the Brazilian Embraer EMB-145I aircraft platform.

The system encompasses a sophisticated suite of mission equipment, including an Active Electronically Scanned Array (AE-SA) radar, Identification Friend or Foe (IFF), mission computer, secure communication networks, electronic support measures (ESM) and communi-



Air Marshal Awadhesh Kumar Bharti and others, pose during a ceremony marking the FOC of NETRA on Thursday. MURALI KUMAR K

cation support measures. India is the fifth country in the world to develop this capability, as Netra can detect, track, identify and monitor airborne and maritime targets enhancing the network-centric operations capabilities of the IAF.

Dedication

At an event to mark the occasion, scientists and IAF officers dedicated the

abandoned temporarily.

The programme took off again after being sanctioned afresh in 2004. During this intervening period, scientists kept their perseverance and hoped for the best.

“On January 11, 1999, the ill-fated aircraft had a crash. I salute personnel who were on board, who made their supreme sacrifice. Today we have shown that their sacrifice has not gone in vain. This FOC is dedicated to them,” Director-General of Aeronautics Cluster of the Defence Research and Development Organisation (DRDO), K. Rajalakshmi Menon said.

The Initial Operational Clearance (IOC) for the Netra was accorded in 2015 and it was inducted into the IAF in 2017.

The Cabinet Committee on Security (CCS) has approved the development of six more AEW&C Mk-1A systems by the CABS.

achievement to colleagues killed in a tragic air crash in 1999. The seeds of India's airborne warning and control system (AWACS) were sown in the early 1980s.

However, the tragic crash of a modified HS-748 Avro aircraft near Arakkonam in Tamil Nadu, with eight people aboard, including four IAF personnel and four scientists, on January 11, 1999 resulted in the programme being

Netra, India's Airborne Surveillance Platform, Gets Final Operational Nod

● **The Development**

- The indigenous **Netra Airborne Early Warning and Control (AEW&C) system** was accorded Final Operational Clearance (FOC) on Thursday
- Netra played a pivotal role as a force multiplier for the IAF in the 2019 Balakot strikes and in Operation Sindoor last year

● **About Netra**

- **Developed indigenously by the Bengaluru-based Centre for Airborne Systems (CABS), under the DRDO**
- Integrated on the **Brazilian Embraer EMB-145I aircraft platform**
- Mission equipment suite includes: Active Electronically Scanned Array (AESA) radar, Identification Friend or Foe (IFF), mission computer, secure communication networks, and Electronic Support Measures (ESM)
- Can detect, track, identify and monitor airborne and maritime targets, enhancing the IAF's network-centric operations capabilities

India is the 5th country in the world to develop this capability

Thank You